# *SMART IRRIGATION SYSTEM USING SOIL MOISTURE SENSOR*

(CHANDIGARH ENGINEERING COLLEGE)

PROJECT – I

## Final report on Smart Irrigation System Using Soil Moisture Sensor:

**Submitted by:**

PRANSHUL SHARMA (2420802)
REHANSH KAMBOJ (2420816)
SWAPNODEEP PRADHAN (2420862)
JASLEEN KUMAR BHATTI (2420735)

**Submitted to:**

Ms. Gagan

## Table of Contents


| Sr. No. | Contents |
|---|---|
| 1 | Introduction |
| 2 | Objectives |
| 3 | Methodology |
| 4 | Applications |
| 5 | Advantages |
| 6 | Expected Outcome |
| 7 | Conclusion |

## Project Title: Smart Irrigation System Using Soil Moisture Sensor

### 1. Introduction

Agriculture is the backbone of every economy and plays a vital role in sustaining human life. However, the challenge of water scarcity and inefficient irrigation practices often leads to reduced crop yields and resource wastage. Traditional irrigation methods involve manual control, which can result in either over-irrigation or under-irrigation.

To overcome this issue, we propose a Smart Irrigation System using a Soil Moisture Sensor that automatically detects the soil's moisture level and controls the water supply accordingly. This system ensures optimum water use, better crop health, and reduced manual effort, making it a sustainable and cost-effective approach for modern agriculture.

### 2. Objectives

- To design an automated irrigation system that reduces water wastage.
- To monitor soil moisture levels in real time and supply water only when required.
- To minimize human intervention in irrigation and improve overall crop productivity.

- To develop a low-cost, efficient, and sustainable solution for farmers and gardeners.
- To promote the use of IoT and smart technology in agricultural systems for better control and efficiency.

## 3. Methodology

A. Hardware Components

- Soil Moisture Sensor: Detects the moisture content in the soil.
- Microcontroller (Arduino/ESP32): Processes sensor data and controls the pump.
- Relay Module: Acts as a switch between the microcontroller and water pump.
- Water Pump / Solenoid Valve: Regulates the water flow to the soil.
- Power Supply: Provides power to all components.
- LCD Display / IoT Dashboard (optional): Displays the soil moisture level and system status in real time.

B. Working Principle

- The soil moisture sensor is placed in the soil to measure the current moisture level.
- The sensor continuously sends data to the microcontroller.

- When the moisture level drops below a defined threshold, the microcontroller activates the water pump through the relay module.
- Water is supplied to the soil until the moisture level reaches the desired value.
- Once the soil has enough moisture, the pump is automatically turned off.
- The system ensures accurate irrigation without any human effort.

## 4. Applications

- Agriculture and Farming: For efficient irrigation of crop fields.
- Home Gardening and Lawns: For maintaining household plants automatically.
- Greenhouses and Nurseries: To maintain ideal conditions for plant growth.
- Smart Cities and Urban Farming: For water-efficient irrigation in limited urban spaces.
- Agri-tech Research: For testing smart farming concepts and sustainability models.

## 5. Advantages

- Saves Water: Supplies only the required amount of water to plants.
- Reduces Human Effort: Eliminates the need for manual monitoring and irrigation.

- Improves Crop Yield: Ensures healthy plant growth by maintaining ideal moisture levels.
- Eco-friendly: Promotes sustainable and responsible water management.
- Cost-effective: Uses low-cost components suitable for small and large-scale applications.
- Scalable: Can be easily expanded using IoT for remote monitoring and control.

## 6. Expected Outcome

- A fully functional prototype of a smart irrigation system that automates water supply based on soil moisture levels.
- Reduced water consumption and improved irrigation efficiency.
- Better crop growth and productivity due to optimal moisture management.
- A sustainable solution adaptable for both small gardens and large farms.
- Potential for integration with IoT for remote operation and real-time data visualization.

## 7. Conclusion

The Smart Irrigation System using Soil Moisture Sensor offers an innovative and sustainable approach to modern farming. By automating the irrigation process, it ensures that crops receive the right amount of water

at the right time. This system helps in conserving water, reducing labor, and increasing productivity, making it highly beneficial for farmers and gardeners alike.

With its cost-effective design and potential for IoT integration, this project represents a significant step toward smart agriculture and sustainable resource management in the future.